CO1 - Assesment Tool 1 : Concept Mapping

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CSAG506 - Gen AI

98/100

1) Evolution from AI to foundation models.

Artificial Intelligence

Goal: make machines perform tasks that require human intelligence

→ **Limitation**
- → Rule based
- → Limited adaptability
- → No learning from data

Machine learning (ML)

Goal: Learn patterns from data without explicit programming

→ **Improved over AI**
- → Handles data
- → Learns from experience
- → Better accuracy

Deep learning (DL)

Goal: use multi-layer neural networks to learn complex patterns.

→ **Improved over ML**
- → Learns hierarchical features
- → Works with large data
- → Higher performance

Generative AI (GenAI)

Goal: Generate new content (text, images, audio, code etc..

→ **Improved over DL**
- → Creates new data
- → Learns data distribution
- → More creative & flexible.

Foundation models (FMs)

Goal: Large, general purpose models trained on massive data for multiple tasks.

→ **Improved over Gen AI**
- → Broad knowledge
- → Transfer learning
- → Multi-modal
- → Few shot capability.

KEY RELATIONSHIPS

AI → ML → DL → Gen AI → Foundation models

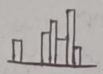
From rule based systems → to data driven learning → to creative generation → to general purpose intelligence

2) Discriminative vs Generative models.

Two types of models.

Discriminative models
Learn the boundary between classes (P(x|y) or directly P(y|x))

Generative models
Learn the data distribution and generate new samples.

	Discriminative models		Generative models
	Probability Distribution. → P(y/x) → Focus on decision boundary	Probability distribution	Probability distribution → P(x) or P(x,y) → models the joint distribution
	Learning objective. → minimize classification error eg., crossentropy, hinge loss	Learning objective	Learning objective → maximize likelihood → eg., BCE, MSE
	Neural networks. → feed forward networks e.g:- CNN, MLP	Neural networks	Neural networks → encoder-decoder, GAN, VAE, transformer (eg GPT)
	Training Process → supervised learning → optimize classification loss	Training Process	Training Process → unsupervised / self-supervised → eg., reconstruction
	Applications. → classification prediction e.g., spam detection, image recognition	Applications	Applications. → new content generation eg., text, image, audio, code.